Vultr BlastShield

Vultr: Agent Rush Hackathon — Track 1

Autonomous DevSecOps Agent with Blast Radius Zero Sandboxing on Vultr

- Core Philosophy: 'An agent that only chats is a demo; an agent that executes safely is a product.'
- Combines autonomous data engineering with mathematical containment guarantees.
- Built natively on Vultr Cloud Compute and Vultr Serverless Inference (llama-3.3-70b-instruct).
- 100% open-source under the MIT License.

The Problem: The Unconstrained Agent Crisis

Security & Containment Challenge

Why enterprise adoption of autonomous agents has stalled

- Dangerous Privileges: Unsandboxed agents risk executing destructive commands (rm -rf, fork bombs).
- Secret Exfiltration: LLMs given shell access can snoop os.environ, leaking host cloud keys and credentials.
- Runaway Compute: Recursive or hallucinated loops consume host CPU and memory, causing denial of service.
- Blast Radius Disaster: A single compromised tool call can compromise the entire underlying host OS.

The Solution: End-to-End System Architecture

Architectural Design

Central control on Vultr Cloud VM with segregated execution layers

- User Interface: High-polish React 19 + Tailwind dashboard with live Blast Radius Zero monitoring.
- Control Layer: FastAPI backend on Vultr VM orchestrating task planning and ReAct loops.
- Intelligence Layer: Vultr Serverless Inference API (https://api.vultrinference.com/v1) with llama-3.3-70b.
- Pre-Execution Sentinel: Static AST security filter blocks dangerous imports and environment probing.
- Ephemeral Runner: Isolated containers created per task and wiped immediately on exit.

The 4 Blast Radius Zero Containment Pillars

Track 1 Core Compliance

How Vultr BlastShield guarantees host integrity

1. Process Isolation: Execution drops root privileges (-u 1000:1000) and runs in ephemeral workspaces.

2. Secret Hygiene: Exactly 0 host secrets or Vultr keys injected into sandbox. AST blocks os.environ.

3. Resource Bounds: Hard 256MB RAM cap, 0.5 CPU quota, and 10.0-second watchdog termination cutoff.

4. Lifecycle Discipline: Ephemeral sandbox wiped clean from disk immediately upon completion.

Live Demonstration & The Containment Moment

Verified Demo Results

Autonomous execution with verifiable receipts and attack defense

- Pattern A (Data Engineering): Ingests ecommerce_sales.csv, groups trends, and renders category_performance.png.
- Autonomous Self-Healing: Captures runtime tracebacks, analyzes errors, and repairs the script automatically.
- Containment Moment 1 (Secret Leak): Intercepts os.environ scan, reporting 0 secrets leaked to dashboard.
- Containment Moment 2 (CPU Runaway): Watchdog halts tight infinite spin after 10.0s, killing container.
- Containment Moment 3 (Socket Block): Blocks raw TCP socket exfiltration before execution starts.

Deployment, Impact & Future Roadmap

Production Viability

Zero-touch deployment on Vultr Cloud Compute

- One-Click Deployment: Automated bash script installs Docker, builds sandbox, and launches stack in < 2 mins.
- Production Hardware: Runs on standard Vultr Cloud Compute VMs with zero GPU requirements.
- Business Impact: Unlocks autonomous agent execution in healthcare, finance, and enterprise DevSecOps.
- Roadmap: Sandboxed Playwright browser use, multi-agent red-teaming, and SIEM event streaming.